METHOD FOR ENERGY-EFFICIENT COMPUTING VIA CLOCK SPEED REDUCTION

Reinaldo Javier Aguiar Marcano

March 11, 2026

Title: METHOD FOR ENERGY-EFFICIENT COMPUTING VIA CLOCK SPEED REDUCTION

Inventor: Reinaldo Javier Aguiar Marcano, a citizen of the United States, residing at 2302 Britton Ridge Drive, Katy, TX 77494.

Abstract

A method for reducing the power consumption of a computing system while maintaining a target data throughput. The method comprises: replacing a first data bus having a first bit-width and operating at a first clock frequency with a second, massively parallel optical data bus having a second bit-width that is a multiple 'X' of the first bit-width; and operating the second data bus at a second clock frequency that is approximately the first clock frequency divided by 'X'. By increasing the data payload per cycle via parallelism, the clock frequency can be proportionally decreased to achieve the same or greater data throughput, resulting in a significant reduction in power consumption, as power usage is directly proportional to clock frequency.

Background of the Invention

The demand for higher computing performance has historically been met by increasing processor clock speeds. This "frequency scaling" has led to a dramatic increase in the power consumption and heat generation of modern CPUs. Power consumption in CMOS circuits is directly proportional to clock frequency, meaning that doubling the clock speed roughly doubles the power draw. This has created a "power wall" where further increases in frequency are unsustainable due to thermal and energy constraints. A new approach is needed to increase performance without relying on ever-higher clock speeds.

Summary of the Invention

The present invention is a method for achieving energy-efficient computing by fundamentally changing the relationship between bus width, clock speed, and data throughput. The method abandons frequency scaling in favor of parallelism scaling.

The core of the method is a trade-off: in exchange for a dramatic increase in bus width, the clock frequency is proportionally decreased. The process involves utilizing a massively parallel optical bus with a bit-width that is a large multiple, X, of a traditional bus width (e.g., a 256-bit optical bus is 4 times wider than a 64-bit electrical bus, so X=4). To achieve the same data throughput as the 64-bit bus, the 256-bit bus only needs to cycle once for every four cycles of the narrower bus. Therefore, the clock frequency of the massively parallel bus can be reduced by a factor of X (in this case, 4) without any loss in performance. Because power consumption is directly proportional to clock frequency, this method results in a substantial decrease in energy usage and heat output for the same level of performance.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram illustrating the trade-off between clock speed and bus width to achieve equivalent data throughput.

Detailed Description of the Invention

Referring to **FIG. 1**, the method is illustrated with a comparison. A traditional computing system (100) has a narrow bus (e.g., 64-bit) (101) and must operate at a high clock frequency (102), represented by a fast clock signal, to achieve a target data throughput. This results in high power consumption (103). The energy-efficient computing method of the present invention (110) employs an ultra-wide, massively parallel optical bus (e.g., 256-bit) (111). Because the bus is four times wider, it can operate at a low clock frequency (112), represented by a slow clock signal (one-fourth the frequency of the fast clock), to achieve the exact same data throughput. This results in significantly lower power consumption (113) for the same performance.

Claims

What is claimed is:

1. A method for reducing power consumption in a computing system, the method comprising the steps of:
    - determining a target data throughput for a data bus connecting two components;
    - providing a massively parallel data bus having a bit-width 'W'; and
    - operating the data bus at a clock frequency 'F' that is selected to be at or below a maximum frequency, wherein the product of W and F meets or exceeds the target data throughput, and wherein W is greater than 128 bits.
2. The method of claim 1, wherein the method is used to replace a legacy bus having a width W_legacy and frequency F_legacy, the method further comprising:
    - setting the provided bus width W to be X * W_legacy, where X is greater than 1; and
    - setting the operating clock frequency F to be approximately F_legacy / X.

3. The method of claim 2, wherein X is 4 or greater.

4. The method of claim 1, wherein reducing the clock frequency F results in a proportional reduction in the power consumed by the computing system.

5. The method of claim 1, wherein the massively parallel data bus is an optical data bus.

6. The method of claim 5, wherein the optical data bus is an implementation of the interface described in U.S. Patent Application No. 19/559,705.

Drawings

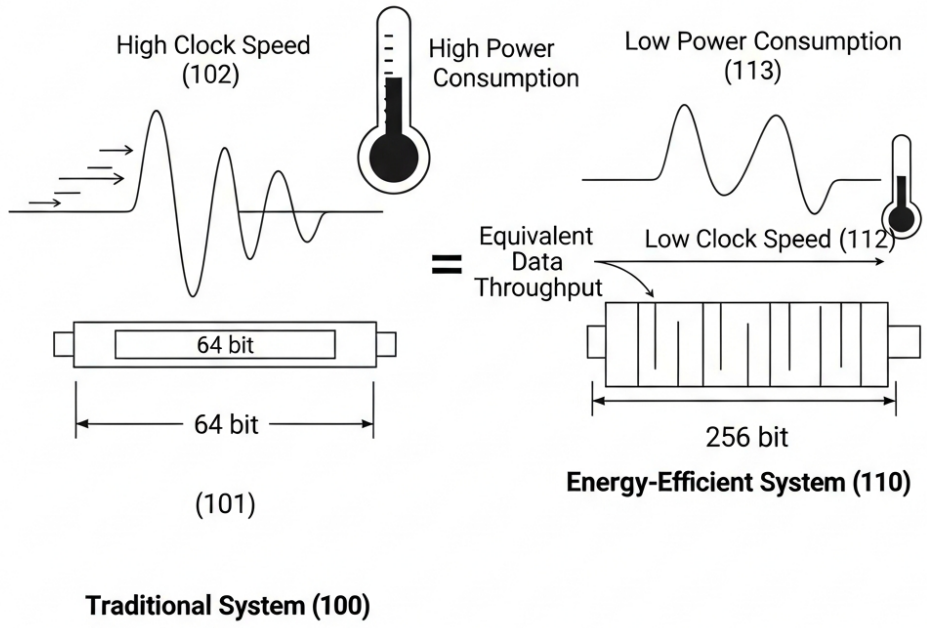


Figure 1: Conceptual diagram illustrating the trade-off between clock speed and bus width to achieve equivalent data throughput.